

Dataset: Long-term LoRaWAN Communication Metadata from an Urban Deployment

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ABSTRACT

This paper presents a dataset containing long-term long-range wide-area network (LoRaWAN) communication metadata collected from a sensor network deployed on the University of Virginia (UVA) campus. It includes transmission records from 10 sensors communicating with three gateways, resulting in 30 unique sensor-gateway metadata pairs. The data spans from February 9, 2024, to June 11, 2025, and includes metrics such as received signal strength indicator (RSSI), signal-to-noise ratio (SNR), spreading factor (SF), frequency, airtime, and more. Additionally, weather data from a co-located weather station is provided to support the environmental context. This dataset aims to address the scarcity of real-world, long-duration LoRaWAN metadata resources, providing researchers with a valuable foundation for studying wireless signal behavior, data-driven communication modeling, and robust low-power WAN (LPWAN) system design.

CCS CONCEPTS

• Computer systems organization → Sensor networks.

KEYWORDS

IoT, LoRaWAN, LoRa, LPWAN, RSSI, SNR, LoS, NLOS

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1 INTRODUCTION

The rapid growth of the Internet of Things (IoT) has driven demand for low-power, long-range communication technologies [8, 19, 23]. Long-range wide-area network (LoRaWAN), a widely adopted low-power WAN (LPWAN) protocol [21], enables energy-efficient and scalable wireless communication in both urban and rural settings. Its reliance on unlicensed spectrum, adaptive data rates, and robust modulation techniques (e.g., chirp spread spectrum) makes it suitable for a diverse range of applications, from environmental monitoring to smart infrastructure.

While LoRaWAN is designed primarily for data transmission, signal-level metadata—such as received signal strength indicator (RSSI), signal-to-noise ratio (SNR), and spreading factor (SF)—also encodes rich information about the radio environment. These physical-layer parameters are influenced by ambient deployment conditions, including obstacles, antenna placement, and weather [11, 17, 18], which affect signal propagation characteristics.

Recognizing the research potential of LoRaWAN metadata, several recent studies have released transmission datasets from their LoRaWAN deployments [3–5, 7, 15, 22]. Such datasets enable applications beyond connectivity, including link reliability analysis, environmental inference from RF signals, and adaptive parameter tuning in deployed LoRaWAN networks.

However, existing publicly available datasets exhibit limitations such as short observation periods, irregular sampling, excessively low sampling rates, or a lack of environmental context. To address these gaps, we present a comprehensive dataset capturing LoRaWAN transmission metadata collected over 16 months (Feb 2024 – Jun 2025) from a real-world sensor network deployed on the University of Virginia (UVA) campus. The deployment includes 10 sensors transmitting data every five minutes and three gateways (indoor and outdoor), resulting in 30 unique communication pairs. To enrich the understanding of the deployment's



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ambient conditions, the dataset also includes weather observations from a co-located station. The dataset is publicly available through LibraData, the UVA's data repository [16] (<https://doi.org/10.18130/V3/RFTICK>).

In addition to data publication, we provide our insights into the behavior of LoRaWAN communication observed in our deployment. Our metadata analysis reveals that:

- Gateway deployment conditions shape signal profiles.
- Sensors deployed indoors yield more stable and stronger signals than outdoors.
- Line-of-Sight (LOS) dominates signal stability, strength, and noise level.

These observations not only highlight real-world deployment challenges but also offer opportunities for further research. Our goal is to support the research community with a valuable resource for studying wireless propagation, inferring environmental context from RF metadata, and designing resilient LoRaWAN networks.

2 RELATED WORK

To contextualize our contribution, Table 1 compares representative LoRaWAN datasets along dimensions such as sampling interval, duration, deployment scale, device mobility, and recorded radio parameters. While most existing datasets feature irregular or short-term collection, our dataset combines periodic transmissions (every 5 minutes), multi-gateway deployment, and 16 months of continuous data from fixed-location sensors, offering a rich resource of long-term and real-world LoRaWAN deployment metadata.

3 TESTBED DEPLOYMENT

LoRaWAN is a low-power, wide-area communication protocol widely adopted for scalable IoT applications. It operates in the unlicensed ISM radio bands using chirp spread spectrum (CSS) modulation, which enables long-range, energy-efficient wireless communication [2]. LoRaWAN also supports secure, bi-directional communication between IoT devices and gateways through adaptive data rate (ADR), optimizing parameters such as SF and transmission power [9]. In a typical LoRaWAN setup, IoT devices periodically transmit encrypted data packets to nearby gateways, which forward them to cloud-based network servers such as The Things Network (TTN) [21].

We deploy a real-world LoRaWAN network across a 590×1120 meter area on the University of Virginia campus, located in a city with a four-season humid subtropical climate. The deployment comprises three Cisco IXM-LPWA-900-16-K9 [6] gateways (two outdoor and one indoor), ten off-the-shelf LoRa-enabled sensors, and a co-located AcuRite Iris weather station [1] for environmental context (Figure 1).

We mount one gateway (gatewayA) on the roof of a five-story office building for wide coverage (Figure 2a), place

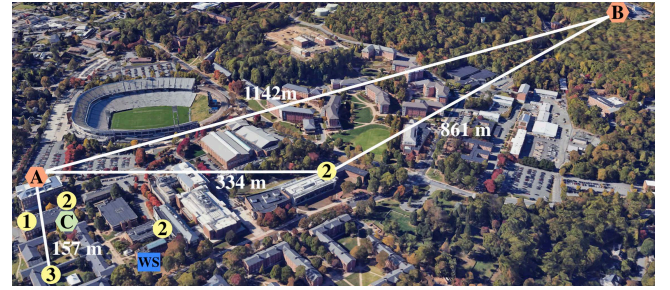


Figure 1: LoRaWAN deployment map showing the locations of three gateways, ten sensors, and a weather station across the UVA campus. Orange hexagons (gateways A and B) represent outdoor gateways, while the green hexagon (gatewayC) indicates the indoor gateway. The yellow circles indicate 10 sensors (each circle represents the number of deployed end-nodes in that area), and a weather station (WS) for ground truth measurements depicted as a blue square.



(a) GatewayA: outdoor rooftop deployment.



(b) GatewayC: indoor second-floor deployment.

Figure 2: Gateways deployment example in both outdoor and indoor settings.

another (gatewayB) on a single-story lab building, and locate the indoor gateway (gatewayC) on the second floor of a two-story office building beside a window (Figure 2b), introducing non-line-of-sight (NLOS) conditions. As end-nodes, we use Linovision S500TH (temperature/humidity) and S500CO2 sensors [13, 14]. We place four pairs of S500TH sensors in four distinct buildings, with each pair consisting of one sensor indoors and one sensor outdoors. In addition to these, we install an S500CO2 sensor indoors on the third floor of a three-story building, and place a final S500TH sensor inside an office on the second floor of a two-story building.

We configure each sensor to transmit data every five minutes. The network operates in 902–928 MHz ISM band with ADR enabled, and each end-node transmits at a nominal power of 20 dBm. The Cisco gateways use the standard LoRa packet forwarder to communicate with TTN servers [21] for uplink and downlink traffic. Our deployment reflects real-world usage, with dynamic urban environmental factors such as dense pedestrian traffic between academic buildings, nearby stadium crowd activity during football games,

Table 1: Representative LoRaWAN datasets compared to our long-term dataset.

Dataset	Sampling	Duration	Gateways / Devices	Devices Mobility	Radio Parameters
Our dataset [16]	Periodic (5 min)	16 months	3 / 10	Fixed	RSSI, SNR, SF, frequency, bandwidth, airtime, and frame counter
Blenn & Kuipers [4]	Irregular	8 months	691 / 1,618	Mixed	RSSI, SNR, SF, frequency, and bandwidth
Cardell-Oliver et al. [5]	Periodic (30 min)	~10 months	2 / 5	Fixed	RSSI and SNR
Bhatia et al. [3]	Irregular	~4 months	9 / many	Mixed	RSSI, SNR, SF, frequency, bandwidth, coding rate, and cyclic redundancy check
Goldoni et al. [7]	Irregular	85 days	1 / 8	Fixed	RSSI and SNR
Moradbeikie et al. [15]	Periodic (10 s)	5 hours	3/3	Mixed	RSSI and SNR
Vargas Villar et al. [22]	Seconds–minutes (drives)	8 hours	— / 1	Mobile	RSSI, SNR, frequency, bandwidth, and data rate

and physical obstructions like trees and low-rise buildings influencing the communication quality.

Unlike software-defined radios (e.g., USRPs), our commercial off-the-shelf setup does not expose raw IQ samples or lower-layer physical signals. Instead, we rely entirely on signal-level metadata, such as RSSI, SNR, SF, frequency, airtime, and others, as provided by the TTN API.

4 DATA COLLECTION METHODOLOGY

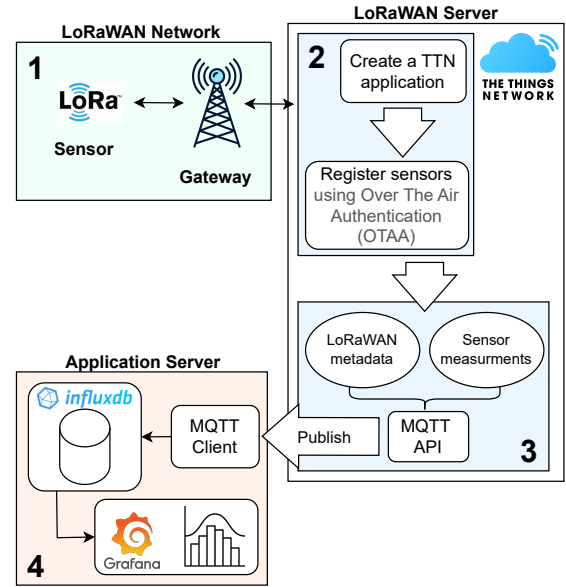
We deploy and connect our LoRaWAN gateways as packet forwarders to TTN server [21] North America cluster (nam1), hereafter referred to as the LoRaWAN network server. This server runs The Things Stack Sandbox [20], a crowd-sourced, open, and decentralized LoRaWAN network intended for non-commercial applications.

The LoRaWAN network server organizes groups of devices into applications, each with its own set of collaborators and integration services. An overview of our data collection workflow is shown in Figure 3. We first create a new application within TTN and register our sensors using over-the-air authentication (OTAA). Each device is assigned a unique set of credentials—AppEUI, DevEUI, and AppKey—which the server uses to authenticate the device and establish its secure connection to the network.

To collect communication metadata, we use TTN’s built-in MQTT integration, which offers a low-overhead and reliable pub/sub communication solution between the network server and our application server. We generate an API key with appropriate access rights to read uplink messages, and connect an MQTT client on our application server to TTN’s MQTT broker using this key. Each time an uplink message is received from a sensor, the LoRaWAN network server decodes its payload, appends transmission metadata (e.g., timestamp, RSSI, SNR, SF), and publishes the enriched message to the MQTT broker.

In our application server, the MQTT client subscribes to the broker, parses the received messages (including both payload and metadata), and writes them to a time-series database.

We use InfluxDB [10], an open-source, high-performance time-series database, for storage. To support visualization and exploratory analysis, we deploy a Grafana [12] instance connected to InfluxDB, enabling real-time inspection, dashboard creation, and data export.

**Figure 3: Data collection process**

5 DATASET

The dataset comprises 2,091,512 LoRaWAN transmissions collected from February 9, 2024 to June 11, 2025, from a real-world testbed deployment on the University of Virginia (UVA) campus. It includes 10 LoRa-enabled sensors transmitting every 5 minutes and three gateways (two outdoor and one indoor), resulting 30 unique sensor–gateway communication pairs. Each transmission record includes signal metadata such as RSSI, SNR, SF, frequency, bandwidth, airtime, and frame counter, described in Table 2. The dataset also includes weather observations from a co-located weather

station for environmental context. The full dataset is publicly available via the UVA Dataverse repository [16].

Table 2: Dataset parameters

Parameter	Unit	Description
Timestamp	milliseconds	Reception timestamp in UTC.
Sensor Alias	—	Unique identifier of the sensor device.
Gateway Alias	—	Unique identifier of the receiving gateway.
RSSI	dBm	Received signal strength.
SNR	dB	Signal-to-noise ratio.
Spreading Factor	—	LoRa modulation spreading factor.
Bandwidth	Hz	Bandwidth used for transmission.
Frequency	Hz	Center frequency used.
Airtime	seconds	Time on air for the packet.
Counter	—	Frame counter from the sensor.
#Receiving Gateways	—	Number of gateways that received the same packet.

We organize the dataset into following folders:

- *dataset/lorawan_metadata/*: Contains 30 individual *.parquet* files, one for each sensor–gateway pair. Each file’s name is in the format of *sensorXX_gatewayY.parquet* (e.g., *sensor01_gatewayA.parquet*). It also includes a combined data file called, *lorawan_combined_dataset.parquet*, which aggregates all transmissions.
- *dataset/summary_statistics/*: Contains computed statistics over the full 16-month data duration. The file name is: *lorawan_summary_by_pair.csv*, which summarizes each sensor–gateway pair with the following metrics: count, min, max, mean, standard deviation (std) for all signal parameters such as RSSI, SNR, SF, and airtime.
- *dataset/weather/*: Includes weather station data in: *deployment_weather.parquet*. This file contains 5-minute interval readings of temperature, humidity, pressure, wind speed, and precipitation we name as *deployment_weather.parquet*. Approximately two months of data are missing due to a temporary sensor outage.

While our dataset provides comprehensive coverage of LoRaWAN transmissions across a variety of real-world deployment conditions, it has a few limitations. First, due to the longer range and more obstructed path to gatewayB, some sensor–gateway pairs, particularly those farther from this gateway, experience intermittent packet loss. Second, sensor08 and sensor09 that we install on the first floor of a building located farther from the rest of the sensors (see Figure 1), experience frequent packet loss when communicating with the indoor gateway (gatewayC). It is likely due

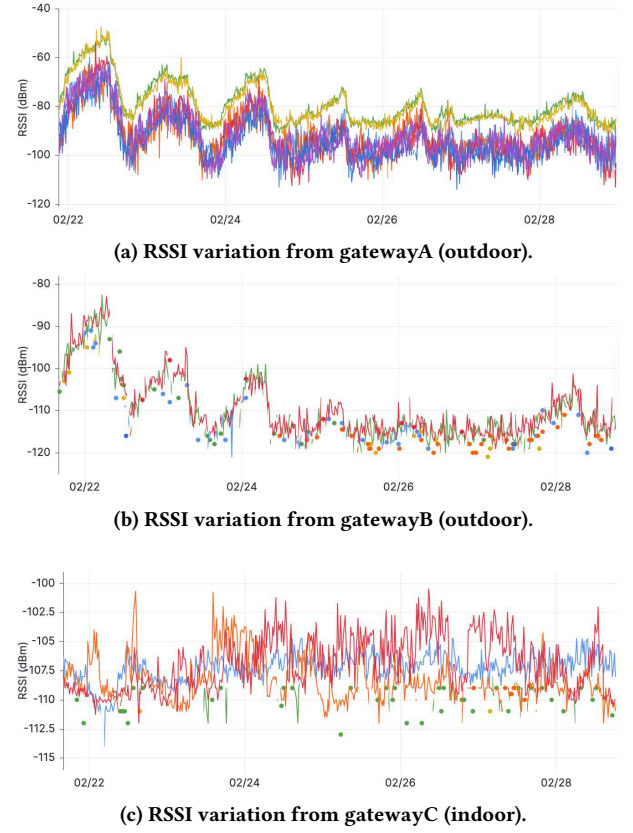


Figure 4: RSSI signal profiles observed at different gateways over time. Each subfigure illustrates the RSSI dynamics shaped by gateway placement and environmental exposure.

to severe NLOS conditions and dense building obstructions along the signal path.

Despite these limitations, such cases reflect realistic operational challenges and provide valuable opportunities for studying signal reliability and network resilience in constrained urban environments.

6 DATASET EMPIRICAL OBSERVATIONS

Our long-term LoRaWAN deployment metadata offers a rich foundation for analyzing network performance, signal quality, environmental inference, and overall system reliability in real-world scenarios. We summarize our three key findings from the dataset below, offering insights that can help researchers and practitioners better understand LoRaWAN propagation characteristics and optimize future deployments.

1) Gateway deployment conditions shape signal profiles: Our data observations indicate that signal characteristics are influenced not only by the presence or absence of LOS, but also by the physical deployment conditions of

the gateways. Specifically, we observe notable differences in signal shape depending on whether a gateway is deployed indoors or outdoors.

The two outdoor gateways (gatewayA and gatewayB) exhibit remarkably similar signal fluctuation patterns over time compared to the indoor gateway (gatewayC). Notably, gatewayA, which is physically closer to most of the sensors, reports a higher average RSSI than gatewayB. Figure 4 compares the three gateways' RSSI profiles over the last week of February 2025. This similarity suggests that environmental conditions, particularly outdoor weather factors such as temperature and humidity, may play a significant role in shaping radio propagation behavior, beyond simple geometric distance.

2) Sensors deployed indoors yield more stable and stronger signals than outdoors: To investigate the effect of sensor placement on signal behavior, we compare two sensor pairs deployed in different buildings: one building with LOS to gatewayA (sensor05 and sensor06), and another building with NLOS to gatewayA (sensor04 and sensor07). In each pair, we install one sensor indoors and the other outdoors. The NLOS sensors are approximately 157 meters away from gatewayA, while the LOS sensors are slightly farther at 196 meters.

Table 3: RSSI and SNR statistics for indoor vs. outdoor sensors reported by gatewayA over the 16-month data collection period

	Sensor ID	RSSI (dBm)		SNR (dB)	
		Mean	Std	Mean	Std
LOS	sensor05 (outdoor)	-83.36	9.99	12.83	3.82
	sensor06 (indoor)	-78.61	9.68	13.67	3.02
NLOS	sensor04 (outdoor)	-94.76	9.94	8.24	5.46
	sensor07 (indoor)	-89.04	9.89	10.86	5.06

A comparison between indoor and outdoor sensors within both LOS and NLOS buildings represented in Table 3, reveals consistent trends of stronger and more stable signal quality for indoor deployments. Specifically, indoor sensors exhibit higher mean RSSI and SNR values, along with slightly lower standard deviations, indicating reduced fluctuation and more reliable link performance compared to their outdoor counterparts.

3) Line-of-Sight (LOS) dominates signal stability, strength, and noise level: We install gatewayA outdoors on the roof of a five-story building, as shown in Figure 2a. Two sensors (sensor05 and sensor06) are deployed inside and outside of a building with a clear LOS to gatewayA. In contrast, all other sensors in the network are in non-LOS (NLOS) positions relative to gatewayA.

Our analysis of reported metadata by gatewayA reveals stronger and more stable RSSI values, with lower signal variability, compared to NLOS configurations. In addition, the average SNR is significantly higher for LOS sensors, highlighting the impact of unobstructed propagation paths on signal clarity and quality.

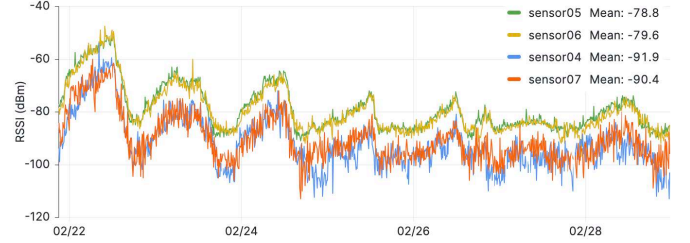


Figure 5: LOS (sensor05 & sensor06) signal attenuation vs NLOS (sensor04 & sensor07)

Figure 5 illustrates the RSSI attenuation patterns during the last week of February 2025, comparing two sensors in a LOS building (sensor05 and sensor06) against two sensors in an NLOS building (sensor04 and sensor07). Although both buildings are approximately the same distance from gatewayA and three of the sensors are positioned at similar heights above ground (sensor05, sensor06, and sensor07) level, the signal strength and stability of the LOS sensors are noticeably superior. Table 4 presents the SNR mean, std, and variance for the four sensors between Feb 22 and Feb 28, 2025.

Table 4: SNR statistics for four sensors (two LOS and two NLOS to gatewayA) during February 22–28, 2025

	Sensor ID	Mean (dB)	Std (dB)	Variance (dB ²)
LOS	sensor05	13.5	2.40	5.74
	sensor06	13.4	2.54	6.47
NLOS	sensor04	9.07	4.49	20.20
	sensor07	9.54	4.29	18.40

7 DISCUSSION

The derived empirical insights from the dataset, which we discuss in Section 6, offer a foundation for several research and engineering applications. For instance, our observations on signal behavior across LOS/NLOS and indoor/outdoor placements can guide more accurate modeling and improve localization accuracy. Furthermore, our analysis reveals that the impact of gateway placement and environmental exposure on signal stability can also inform gateway deployment strategies and adaptive parameter tuning. Additionally, the co-located weather data, combined with our signal analysis, enables investigation into environmental impacts on link quality, supporting more resilient and energy-efficient network designs.

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